College of Engineering

MEE 9995: Master's Project

A Low-Cost Quadruped Robot Dog for K-12 STEM Education

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Introduction

A quadruped robot dog, Squeaky, was developed to address a significant gap in K-12 education: the lack of affordable, durable and approachable robotic platforms. Commercially available robot dogs often cost between $600 and $2000, making them far beyond the budget of most schools, after-school programs, and families. As Program-Director at Coco Academy, a makerspace for kids based in South Philadelphia, the need for low-cost hands-on robotic tools became clear. Squeaky was created to meet this need, providing both financial accessibility and entry level capabilities for students of all ages. To achieve this goal, the project was designed around four key requirements: maintaining a total cost under $100, ensuring durability under rough handling, supporting easy assembly and repair, and incorporating an intuitive custom block-based interface

Mechanical Design

The mechanical design of the robot dog focused on strength and ease of repair. The robot used an eight-degree-of-freedom configuration with two MG996R high-torque servos attached to each leg to control the hip and knee joints.

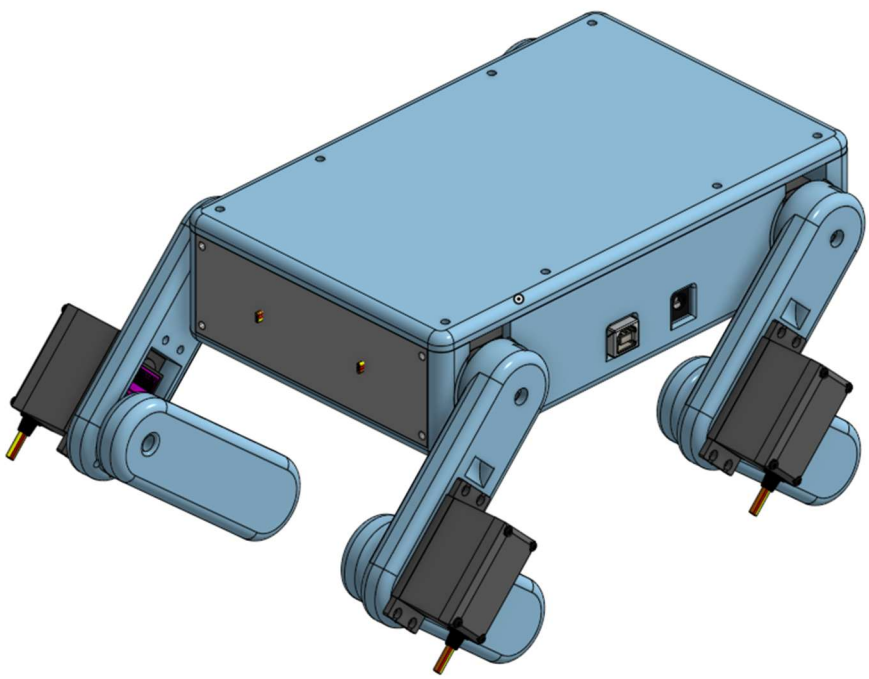


Figure 1: A CAD Model of the Robot Dog

An embedded servo leg architecture was chosen over a linkage-based leg design because it reduces assembly complexity and simplifies the inverse kinematics in the movement logic. All structural parts were designed in Onshape, a browser-based CAD application, and 3D-printed in PLA due to its low cost and adequate stiffness. Layer lines were intentionally oriented to maximize strength and aluminum alloy servo attachments were added to the joints to ensure a strong connection. To support ease of assembly and repair, the robot uses only two screw types and features modular components that can be easily replaced or customized.

Electrical Design

The electrical system was created to focus on real-time responsiveness, low cost and simple integration. The Arduino Uno R4 Wi-Fi microcontroller was selected as the “brain” of the robot due to its built-in ESP32-S3 wireless module and its affordability compared to Raspberry Pi alternatives.

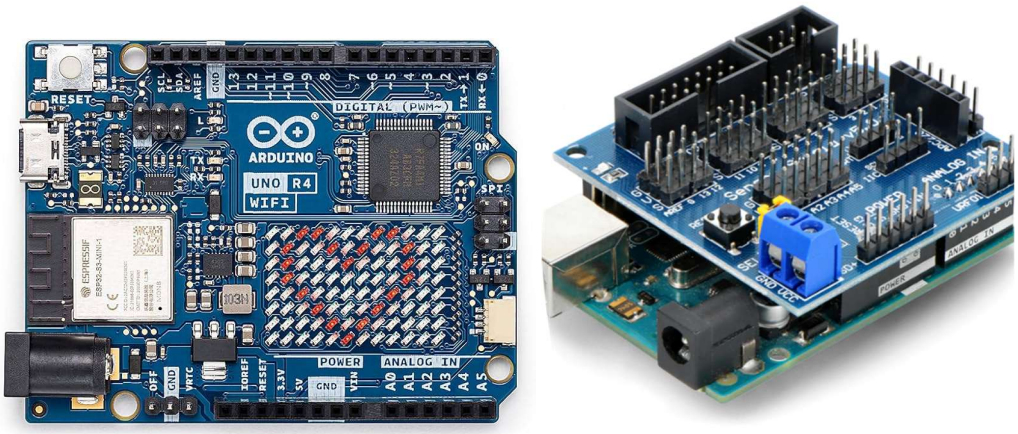


Figure 2: A picture of the Arduino and breakboard used in the robot

The microcontroller is attached to a V5 sensor shield which is used as a breakout board to easily attach all of the servos without complex wiring. Servos were chosen over DC or stepper motors because they include internal potentiometer for position feedback and consume less power. The electrical system is powered by a tethered power supply either to a PC or a wall outlet which allows for easy debugging and no time requirement for charging. The configuration results in a simple but robust system that provides a wireless connection while keeping cost to a minimum.

Durability Testing

Durability was a central requirement throughout the design process as the robot must be able to withstand impacts, drops and rough handling to allow for hours of endless learning. Since most of the frame is 3D-printed, the structural integrity needs to be validated. Finite Element Analysis (FEA) studies were conducted in SimScale on the upper and lower portions of the leg as they are the most likely to be pulled or struck during use. In each study, two trials were made by changing the force from 50N (~11lbs) to 100N (~22lbs). From each trial, the highest Von Mises stress is found and using the Yield Strength of PLA, 40 MPa, the factor of safety is found. The first study conducted focused on the upper leg simulating a student squeezing the legs inwards.

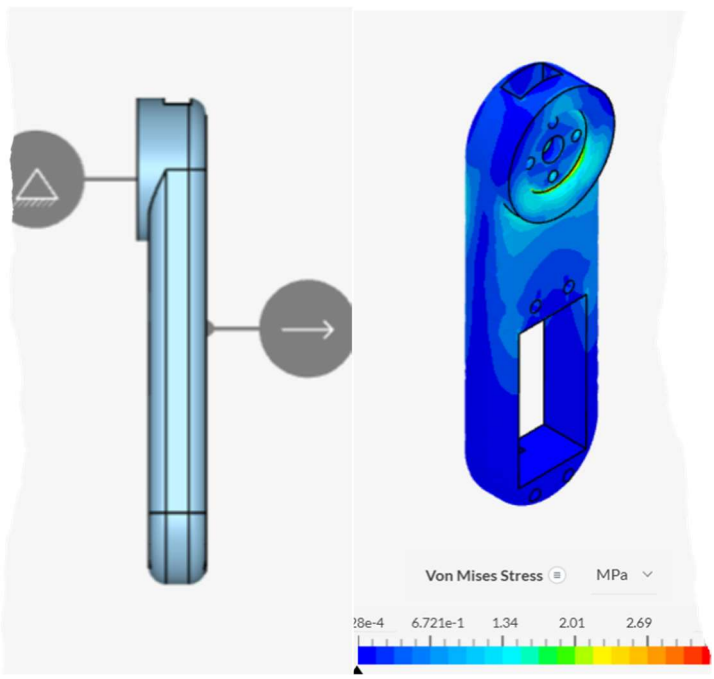


Figure 3: Upper Leg, Distributed Load, FEA Model and Results Plot

A fixed support was made at the servo connection, and the force was applied towards the center of the dog on the outer face. The factor of safety from each trial was 11.9 and 5.96, respectively. The next study also focused on the upper leg but had a point force directed at the bottom of the part simulating bumping into a wall or falling off a table.

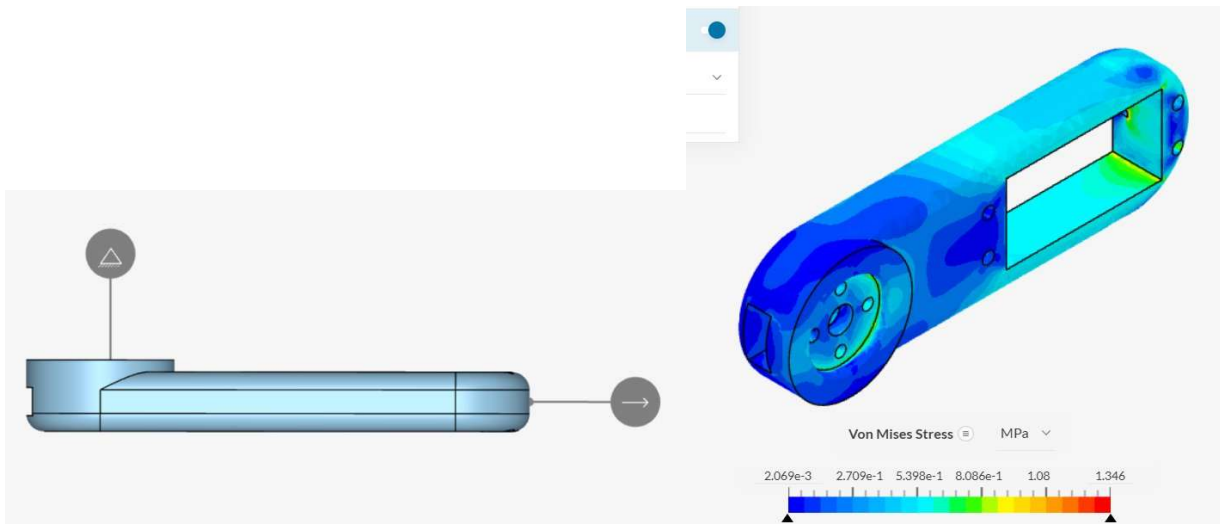


Figure 4: Upper Leg. Point load, FEA Model and Results Plot

A fixed support was made at the servo connection, and the forces were applied as a point load towards the servo connection at the bottom of the part. The factor of safety from each trial was 29.7 and 14.9, respectively. The third study focused on the bottom lab with a distributed load along the inner face simulating a study picking up the dog by the leg.

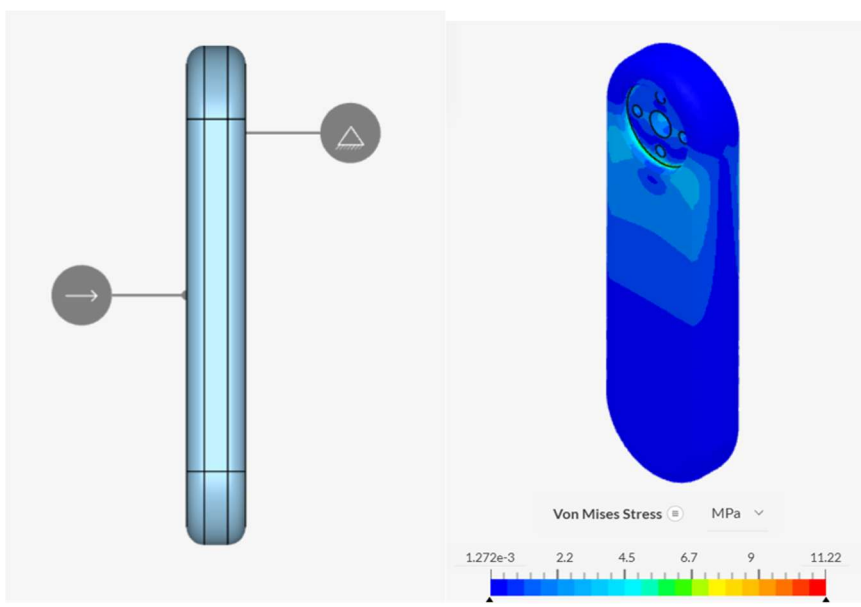


Figure 5: Lower Leg, Distributed Load, FEA Model and Results Plot

A fixed support was made at the servo connection, and the forces were applied outwards to the inner face of the leg. The factor of safety for each trial was 3.56 and 1.78, respectively. Since the factor of safety is lower than 2, there could be some structural integrity issues that arise; however, since the force applied, 100N, exceeds that of normal operational use, it is safe to assume that the leg will not break under normal use. The last

study focused on the lower leg but with a point contact at the bottom of the leg to simulate the robot being dropped or falling off a table.

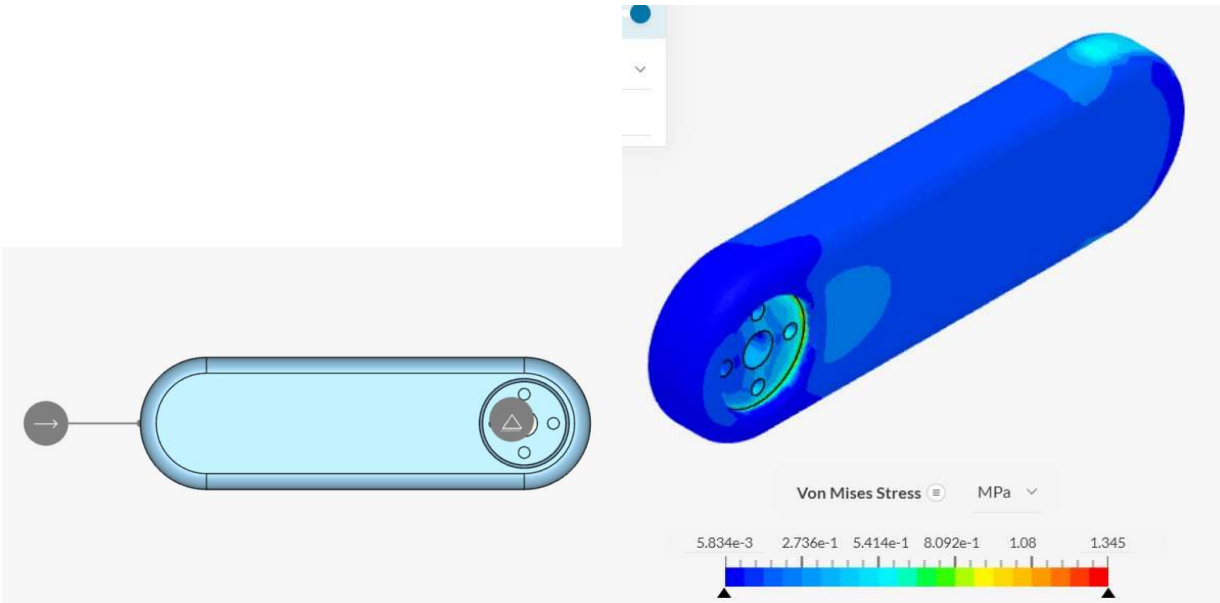
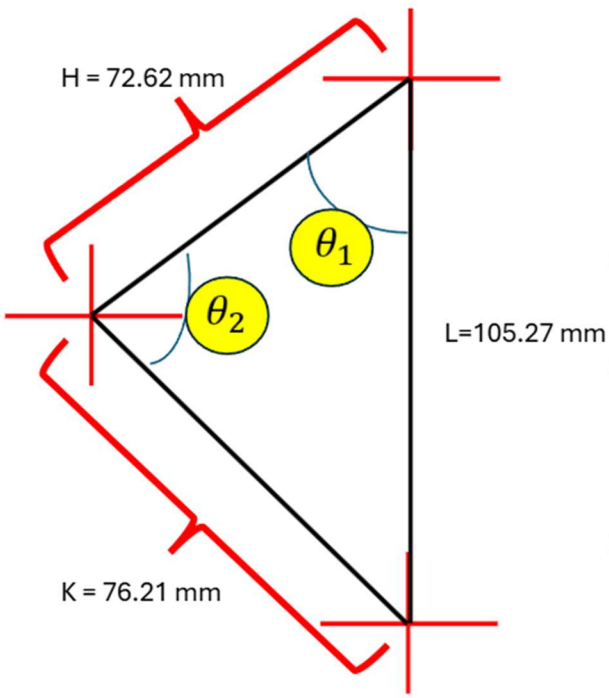


Figure 6: Lower Leg, Point Contact, FEA Model and Results Plot

A fixed support was placed at the servo connection, and a point contact was applied towards the connection from the bottom of the leg. The factor of safety for each trial was 29.7 and 14.9, respectively. Across all simulations, the minimum factor of safety under extreme conditions remained 1.78, confirming that the 3D-printed structure provides sufficient durability for everyday classroom use.

Control Logic

The control logic uses a set of inverse kinematic equations to control the precise movement of the servos. First, the robot dog must undergo a calibration procedure in which each servo is set to 90 degrees. Each servo must be adjusted until each limb is perpendicular to each other. The adjusted neutral position is saved as a calibration offset and later applied to all computer angles to ensure consistency in each step. The leg mechanism is modeled as a two-link planar system and uses the known leg dimensions in the inverse kinematics equations.



Solve for L

- $L = \sqrt{72.62^2 + 76.21^2}$=105.27mm

Solve for θ_2 - Law of Cosines

- $L^2 = K^2 + H^2 - 2(H)(K)\cos(\theta_2)$
- $\boldsymbol{\theta_2 = \arccos\left(\frac{K^2+H^2-L^2}{2(H)(K)}\right)}$

Solve for $\theta_1 - Law\ of\ Cosines$

- $K^2 = L^2 + H^2 - 2(H)(L)\cos(\theta_1)$
- $\boldsymbol{\theta_1 = \arccos\left(\frac{L^2+H^2-K^2}{2(H)(L)}\right)}$

Figure 7: The Inverse Kinematic Calculations for the Leg model.

The inverse kinematic equations are implemented in C++ to compute the target angles for each joint based on a desired foot position in the X-Y space.

```
// ===== Inverse Kinematics =====
void computeIK(float x, float z, float L1, float L2, int &hipIK, int &kneeIK) {
  float L = sqrtf(x*x + z*z);
  float Lmin = fabsf(L1 - L2) + 10.0f;
  if (L < Lmin) L = Lmin;
  float Lmax = L1 + L2 - 1.0f;
  if (L > Lmax) L = Lmax;

  float c2 = clamp1((L1*L1 + L*L - L2*L2) / (2.0f * L1 * L));
  float cK = clamp1((L1*L1 + L2*L2 - L*L) / (2.0f * L1 * L2));

  float a1 = atan2f(-z, x);
  float a2 = acosf(c2);
  float a3 = acosf(cK);

  hipIK  = (int)roundf((a1 + a2) * 180.0f / PI);
  kneeIK = (int)roundf(a3 * 180.0f / PI);
}
```

Figure 8: Implemented Inverse Kinematics

Once computed, the Arduino applies the calibration offset and sends a final angle command to the servos. This approach enables smooth leg motion while keeping computation demand low enough for the microcontroller.

User Interface

To make the robot dog accessible for younger students, a custom drag-and-drop interface was developed using a lightweight web-based architecture. The ESP32-S3 module on the Arduino hosts a small HTTP server that serves an HTML-based control dashboard stored directly in flash memory. The user connects to the robot over Wi-Fi and can interact with a scratch-style coding interface that allows them to control motion without text-based code. The interface includes movement commands such as forward, backward, turning, sitting, standing, as well as logical operators like loops and wait functions.

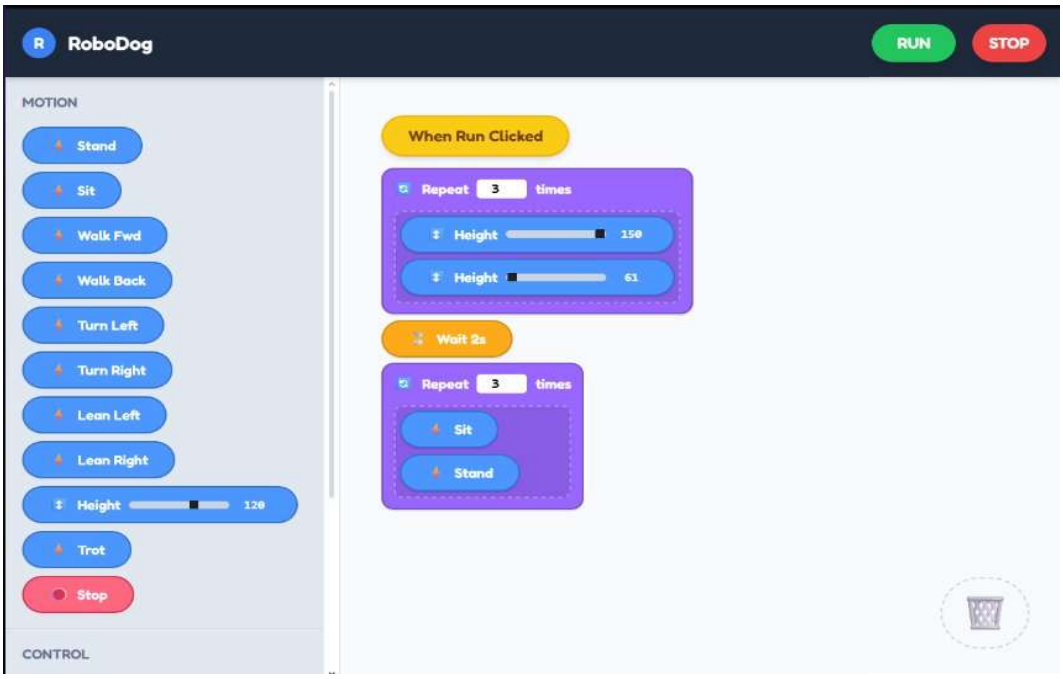


Figure 9: Custom Block-Based Interface

This design prioritizes conceptual understanding of coding and algorithmic thinking which reduces frustration associated with syntax errors. Students can rapidly prototype sequences of movements and see immediate results, engaging and gamifying the learning experience.

Cost Analysis

A major design constraint was keeping the total material costs under $100. Through careful selection of parts and using 3D-printed parts for the majority of the structure, the bill of materials totals approximately $103 when sourced through major retail channels such as Amazon.

Part	Quantity	Price	% of Project Cost
Arduino	1	$29.15	28%
Power Supply	1	$7.84	7%
V5 Sensor Shield	1	$2.83	3%
MG996R Servos	8	$32.84	32%
Aluminum Servo Attachments	8	$14.82	14%
3D Print Material (PLA)	276 grams	$8.28	8%
M3 Screws	1 Pack	$7.99	8%
Total		$103.75	

Costs can be further reduced through sourcing parts from wholesale suppliers such as AliExpress or Alibaba which makes the sub-$100 target practical for bulk production. To consider other potential costs during production such as manufacturing time, assembly and packaging, a 2.5x markup can be applied to achieve a market comparison. The value of a 2.5x markup was found to be a industry guideline for hardware products. Under this model, a reasonable retail price would be approximately $260 which is still far below competing quadruped kits on the market such as Petoi Bittle ($329) and Mangdang Minipupper ($604). This cost structure positions this robot dog as one of the most affordable robotic platforms available for K-12 education.

Conclusion

The quadruped robot dog successfully meets all the key requirements established at the start of the project: affordability, durability, ease of assembly and repair, and compatibility with a drag-and-drop coding system. By keeping the total material cost near the $100 target, the design addresses the need for financially accessible robotic platforms in K-12 education. Durability through FEA demonstrated that the 3D-printed components will hold up under rough handling from younger students. The modular mechanical structure, simple screw selection, and easily replaceable parts ensure that users can maintain the robot with minimal downtime. Lastly, the implementation of a wireless, self-contained, block-based programming interface allows control of the robot without the frustration of

traditional syntax-based coding. With future improvements such as enhanced walking algorithms, additional sensors, and expanded coding features, the robot dog has strong potential to grow into a widely adopted educational tool that helps young students develop engineering intuition and computational thinking skills.